

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (Civil) (Credit System) Degree Examinations
January 2023

20CVE501 - ALTERNATIVE BUILDING MATERIALS AND TECHNOLOGIES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Explain the concept of green design in buildings and its rating.	10	L*2	1	1
	b)	Discuss any four environmental issues concerned to building materials.	10	L1	2	1
2.	a)	What is rainwater harvesting? Discuss the importance of rainwater harvesting. Explain any one technique in rainwater harvesting.	10	L2	2	1
	b)	What is GRIHA? What are the objectives of GRIHA?	10	L1	2	1
3.	a)	Explain any two cost effective building technologies in detail.	10	L1	1	1
	b)	Explain the rating procedure of green building through GRIHA assessment.	10	L2	2	2
		Unit – II				
4.	a)	What are the different types of fibres used in fibre reinforced concrete? Explain any 2 fibres in detail indicating its engineering properties.	10	L2	3	1
	b)	Discuss the importance of use of agro wastes in building constructions. Explain any one agrowaste in detail.	10	L2	4	1
5.	a)	List the different methods of construction of ferrocement. Explain any one method in detail.	10	L2	3	1
	b)	What are Fal G blocks? What are the materials used for FAL-G blocks? Explain the manufactures procedure for FAL-G blocks.	10	L2	4	2
6.	a)	List out the different types of industrial wastes used in building construction. Explain the properties of any two types.	10	L2	3	1,2
	b)	What are stabilized mud blocks? Explain the materials used and properties of stabilized mud blocks.	10	L2	4	1
		Unit – III				
7.	a)	Discuss the importance of use of arches in foundations.	10	L1	5	1,2
	b)	What is Mivar Technology? What are the advantages and disadvantages of Mivar Technology?	10	L2	5	1
8.	a)	Explain the following bridge construction methods. i) Balanced cantilever method ii) Incremental launching method	10	L2	5	1
	b)	Write brief notes on: i) Waffle Slab construction ii) Composites beam panel roofs	10	L1	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
